Implicit Regularization in Deep Linear Networks

Exercise Session — 1h30, pen and paper

References: - Saxe, McClelland & Ganguli, *Exact solutions to the nonlinear dynamics of learning in deep linear neural networks*, ICLR 2014 / PNAS 2019 - Achour, Malgouyres & Gerchinovitz, *The loss landscape of deep linear neural networks: a second-order analysis*, JMLR 2024 - Tu, Aranguri & Jacot, *Mixed Dynamics In Linear Networks: Unifying the Lazy and Active Regimes*, 2024

Setup and notation

We study deep linear networks (DLNs) with L weight matrices $W_1 \in \mathbb{R}^{d_1 \times d_0}, W_2 \in \mathbb{R}^{d_2 \times d_1}, \ldots, W_L \in \mathbb{R}^{d_L \times d_{L-1}}$. The network computes:

$$f(x) = W_L W_{L-1} \cdots W_1 \, x =: W x$$

where $W = W_L \cdots W_1 \in \mathbb{R}^{d_L \times d_0}$ is the end-to-end (or "student") matrix. We train on a dataset $\{(x_\mu, y_\mu)\}_{\mu=1}^N$ with the squared loss:

$$\mathcal{L}(\theta) = \frac{1}{2N} \sum_{\mu=1}^{N} \|y_\mu - W_L \cdots W_1 \, x_\mu\|^2$$

In the population limit with whitened inputs $\Sigma_X = I$, this becomes:

$$\mathcal{L}(W) = \frac{1}{2} \|M - W\|_F^2$$

where $M = \Sigma_{YX} \Sigma_X^{-1}$ is the "teacher" matrix (the OLS solution) with SVD $M = U \operatorname{diag}(s_1, \ldots, s_r, 0, \ldots, 0) \, V^\top$, and $s_1 \geq s_2 \geq \cdots \geq s_r > 0$.

Problem 1 — Loss Landscape Geometry (15 min)

This problem explores the critical point structure of deep linear networks, following Achour et al. (2024).

(a) Consider the **diagonal** case: $d_0 = d_1 = \cdots = d_L = d$, and the teacher is diagonal, $M = \operatorname{diag}(s_1, \ldots, s_d)$, with $s_1 > s_2 > \cdots > s_d > 0$. Restrict attention to diagonal weight matrices $W_l = \operatorname{diag}(w_l^{(1)}, \ldots, w_l^{(d)})$. Show that the loss decomposes into d independent scalar problems:

$$\mathcal{L} = \frac{1}{2} \sum_{\alpha=1}^{d} \left(s_\alpha - \prod_{l=1}^{L} w_l^{(\alpha)} \right)^2$$

(b) For a single scalar mode with target $s > 0$, find all first-order critical points of $\ell(w_1, w_2) = \frac{1}{2}(s - w_1 w_2)^2$ at depth $L = 2$. Show that the critical points are: - The global minimum manifold: $w_1 w_2 = s$ - The origin: $w_1 = w_2 = 0$

Classify the origin as a saddle point by computing the Hessian H of ℓ at $(0, 0)$ and showing it has both positive and negative eigenvalues.

Hint: The gradient equations are $\frac{\partial \ell}{\partial w_1} = -(s - w_1 w_2)\, w_2 = 0$ and $\frac{\partial \ell}{\partial w_2} = -(s - w_1 w_2)\, w_1 = 0$.

(c) Now consider the full (non-diagonal) DLN with $d_l \geq d_L$ for all l. Achour et al. show that every first-order critical point $\theta = (W_1, \ldots, W_L)$ satisfies: there exists a subset $S \subseteq \{1, \ldots, d\}$ such that

$$W = W_L \cdots W_1 = P_S\, M$$

where $P_S = U_S U_S^\top$ is the orthogonal projector onto the span of the left singular vectors $\{u_\alpha\}_{\alpha \in S}$ of M. For the diagonal case $M = \mathrm{diag}(s_1, \ldots, s_d)$, write the value of the loss at the critical point corresponding to $S = \{1, \ldots, r\}$ with $r < d$.

(d) The group $GL_h := GL_{d_1} \times \cdots \times GL_{d_{L-1}}$ acts on the weights by:

$$(W_1, \ldots, W_L) \mapsto (g_1 W_1,\ g_2 W_2 g_1^{-1},\ \ldots,\ W_L g_{L-1}^{-1})$$

Verify that the student map $\mu(\theta) = W_L \cdots W_1$ is invariant under this action: $\mu(g \cdot \theta) = \mu(\theta)$. What does this imply about the dimension of the set of global minima in parameter space?

Problem 2 — Gradient Flow and Conserved Quantities (20 min)

We now study gradient flow: $\dot{\theta}(t) = -\nabla_\theta \mathcal{L}(\theta(t))$, the continuous-time limit of gradient descent with infinitesimal learning rate.

(a) For a two-layer DLN ($L = 2$) with loss $\mathcal{L} = \frac{1}{2}\|M - W_2 W_1\|_F^2$ and whitened inputs, derive the gradient flow equations for each layer. Show that:

$$\dot{W}_1 = W_2^\top (M - W_2 W_1), \qquad \dot{W}_2 = (M - W_2 W_1) W_1^\top$$

Hint: Use the matrix identity $\frac{\partial}{\partial A}\mathrm{Tr}(BAC) = B^\top C^\top$ and the chain rule. Recall $\|X\|_F^2 = \mathrm{Tr}(X^\top X)$. If the matrix calculus is too involved, you may first derive the result for diagonal weight matrices and then state it for the general case.

(b) Define the **balancedness matrix**:

$$G := W_2^\top W_2 - W_1 W_1^\top$$

Show that G is conserved under the gradient flow, i.e. $\dot{G} = 0$.

Hint: Compute $\dot{G} = \dot{W}_2^\top W_2 + W_2^\top \dot{W}_2 - \dot{W}_1 W_1^\top - W_1 \dot{W}_1^\top$, substitute the gradient flow equations from (a), and verify that the terms cancel pairwise.

(c) From now on, assume **balanced initialization**: $G(0) = 0$, which by part (b) means $W_2^\top W_2 = W_1 W_1^\top$ for all time. We want to derive the gradient flow in function space (the ODE for the student $W = W_2 W_1$). This requires several steps.

(c.i) Compute $\dot{W} := \frac{d}{dt}(W_2 W_1)$ using the gradient flow equations from (a). Show that:

$$\dot{W} = (M - W)\, W_1^\top W_1 + W_2 W_2^\top\, (M - W)$$

(c.ii) We now need to express $W_1^\top W_1$ and $W_2 W_2^\top$ in terms of $W = W_2 W_1$. For simplicity, assume that the weight matrices W_l are diagonal. Show that:

$$W_1^\top W_1 = (W^\top W)^{1/2}$$

(c.iii) Similarly, show that $W_2 W_2^\top = (W W^\top)^{1/2}$.

(c.iv) Substitute the results of (c.ii) and (c.iii) into (c.i) to obtain:

$$\dot{W} = (W W^\top)^{1/2}(M - W) + (M - W)(W^\top W)^{1/2}$$

(d) Define the NTK operator for $L = 2$ as:

$$K[F] := (W W^\top)^{1/2}\, F + F\, (W^\top W)^{1/2}$$

Show that the gradient flow from (c) can be written as $\dot{W} = K[M - W]$.

It turns out that this result generalizes to depth L on the balanced manifold and to non-diagonal weight matrices:

$$\dot{W} = \sum_{k=1}^{L} (W W^\top)^{\frac{L-k}{L}} (M - W)(W^\top W)^{\frac{k-1}{L}}$$

Problem 3 — Rich Regime: Incremental Learning (25 min)

This is the core problem. We derive the exact solution of the rich regime dynamics directly from the self-consistent equation of Problem 2, emphasizing the NTK perspective.

(a) Alignment assumption. Start from the balanced gradient flow equation derived in Problem 2(c):

$$\dot{W} = (WW^\top)^{1/2}(M - W) + (M - W)(W^\top W)^{1/2}$$

We work in the **rich regime** (small initialization) with balanced weights. Assume that the student is **aligned** to the teacher: the singular vectors of $W(t)$ coincide with those of M at all times. Concretely, write:

$$M = U\,\mathrm{diag}(s_1, \ldots, s_d)\,V^\top, \qquad W(t) = U\,\mathrm{diag}(w_1(t), \ldots, w_d(t))\,V^\top$$

where U, V are the left/right singular vectors of M, and $w_\alpha(t) \geq 0$ are the evolving singular values of W.

(a.i) Under this alignment assumption, show that the NTK operator is aligned to the task, i.e. $K[F]$ preserves the SVD basis. In particular, show that:

$$(WW^\top)^{1/2} = U\,\mathrm{diag}(w_1, \ldots, w_d)\,U^\top, \qquad (W^\top W)^{1/2} = V\,\mathrm{diag}(w_1, \ldots, w_d)\,V^\top$$

and that the residual is $M - W = U\,\mathrm{diag}(s_1 - w_1, \ldots, s_d - w_d)\,V^\top$.

Hint: Recall that for $A = U\,\mathrm{diag}(\sigma_i)\,V^\top$, we have $AA^\top = U\,\mathrm{diag}(\sigma_i^2)\,U^\top$, and therefore $(AA^\top)^{1/2} = U\,\mathrm{diag}(|\sigma_i|)\,U^\top$.

(a.ii) Substitute into the self-consistent equation. Show that the matrix equation decouples into d independent scalar ODEs:

$$\dot{w}_\alpha = 2\,w_\alpha\,(s_\alpha - w_\alpha), \qquad \alpha = 1, \ldots, d$$

Hint: Compute the product $(WW^\top)^{1/2}(M - W)$ in the SVD basis. It is diagonal with entries $w_\alpha(s_\alpha - w_\alpha)$. The second term contributes identically, giving the factor of 2.

(b) Timescale of learning. The ODE $\dot{w} = 2w(s - w)$ is a logistic equation whose solution is:

$$w(t) = \frac{s}{1 + \left(\frac{s}{w_0} - 1\right) e^{-2st}}$$

where $w_0 = w(0)$. Compute the time t_α it takes for mode α to travel from initial strength w_0 to a final strength w_f. Show that:

$$t_\alpha = \frac{1}{2s_\alpha} \ln\left(\frac{w_f\,(s_\alpha - w_0)}{w_0\,(s_\alpha - w_f)}\right)$$

Deduce that modes with larger singular values s_α are learned faster. This is a **separation of timescales**: the network learns features in decreasing order of their singular value strength.

(c) Incremental learning. Consider a teacher with r nonzero singular values and a small uniform initialization $w_\alpha(0) = w_0 \ll s_r$ for all α. Discuss:

- What is the approximate rank of $W(t)$ at early times?
- How does the rank evolve at intermediate times?
- What happens in the limit $t \to \infty$?

What does this tell us about the sequence of critical points visited by the gradient flow?

Problem 4 — Lazy Regime (15 min)

We now show that large initialization freezes the NTK and eliminates the timescale separation found in Problem 3. For clarity, we work with the **diagonal scalar** model from Problem 1(a), so each mode α is independent. This avoids matrix algebra and isolates the essential mechanism.

(a) Setup. Consider a single mode: a depth-2 diagonal DLN with scalar weights a, b learning a target $s > 0$. From Problems 2–3, the balanced gradient flow for $w = ab$ is:

$$\dot{w} = 2w(s - w)$$

The factor $2w$ is the (scalar) NTK — it is **state-dependent**: the effective learning rate depends on the current value of w.

Now initialize at a **large** value $w_0 \gg s > 0$. Show that in the early phase of training, while w has not changed much from w_0, the ODE becomes approximately:

$$\dot{w} \approx 2w_0(s - w)$$

(b) Frozen-NTK solution. Solve the linearized ODE from (a). Show that:

$$w(t) = s + (w_0 - s)\, e^{-2w_0 t}$$

What is the learning timescale?

(c) No timescale separation. Now restore the mode index α. In the lazy regime, each mode satisfies $\dot{w}_\alpha \approx 2w_0(s_\alpha - w_\alpha)$ with the **same** rate $2w_0$ for all α. Compare the lazy learning time t_α^{lazy} with the rich-regime timescale from Problem 3(b): $t_\alpha^{\text{rich}} \propto 1/s_\alpha$.

Discuss the difference in **rank bias** between the lazy and rich regimes.

Problem 5 — Mixed Dynamics: Unifying Lazy and Rich (15 min)

In practice, networks are neither purely lazy nor purely rich. Tu, Aranguri & Jacot (2024) provide a unified description. We develop the key ideas using the diagonal scalar model.

(a) The interpolating ODE. In Problems 3 and 4, we studied the same ODE $\dot{w}_\alpha = 2w_\alpha(s_\alpha - w_\alpha)$ in two limits: small w_0 (rich) and large w_0 (lazy). A more general model for a depth-2 DLN with balanced initialization at scale σ and width n replaces the scalar NTK $2w_\alpha$ by:

$$\dot{w}_\alpha = 2\sqrt{w_\alpha^2 + \tau^2}\,(s_\alpha - w_\alpha)$$

where $\tau := \sigma^2\sqrt{n}$ is a **threshold** parameter that depends on the initialization scale and width.

Verify that this ODE reproduces the two known regimes: - **Rich limit** ($\tau \to 0$): recover $\dot{w}_\alpha = 2|w_\alpha|(s_\alpha - w_\alpha)$. - **Lazy limit** ($\tau \to \infty$ with w_α bounded): recover $\dot{w}_\alpha \approx 2\tau(s_\alpha - w_\alpha)$.

(b) Two-phase dynamics. At initialization, all modes start at $w_\alpha(0) \sim \sigma^2 \ll \tau$ (since $\sqrt{n} \gg 1$). Observe that: - When $|w_\alpha| \ll \tau$, the effective learning rate is $\approx 2\tau$, the same for all modes (lazy behavior). - When $|w_\alpha| \gg \tau$, the effective learning rate is $\approx 2|w_\alpha|$, which is mode-dependent (rich behavior).

Describe the resulting two-phase dynamics in words: what happens first, and what happens later?

(c) Mode-by-mode transition. Not all modes cross the threshold τ at the same time: which modes cross it first? Explain why this means that a single network can simultaneously have some modes in the rich regime and others still in the lazy regime.

Problem 6 (Bonus) — Stochastic Implicit Bias (10 min)

SGD introduces noise from mini-batching. A continuous model is the Langevin SDE:

$$d\theta_t = -\nabla \mathcal{L}(\theta_t)\, dt + \sqrt{\eta\, \Sigma(\theta_t)}\, dB_t$$

where $\Sigma(\theta)$ is the covariance of the stochastic gradient noise and η is the learning rate.

(a) The probability density $p(\theta, t)$ of the parameters evolves according to the Fokker–Planck equation:

$$\partial_t p = -\nabla \cdot \mathbf{j}, \qquad \mathbf{j} = -\nabla \mathcal{L}(\theta)\, p(\theta) - \frac{\eta}{2} \nabla \cdot (\Sigma(\theta)\, p(\theta))$$

where $\mathbf{j}$ is the probability current. Assume: (i) stationarity $\partial_t p^* = 0$, (ii) thermal equilibrium $\mathbf{j} = 0$, and (iii) isotropic noise $\Sigma = \sigma^2 I$. Show that the equilibrium distribution is the Boltzmann distribution:

$$p^*(\theta) \propto \exp\left(-\frac{2}{\eta\sigma^2} \mathcal{L}(\theta)\right)$$

Hint: Setting $\mathbf{j} = 0$ with $\Sigma = \sigma^2 I$ gives $\nabla \mathcal{L}\, p + \frac{\eta\sigma^2}{2} \nabla p = 0$. This is a first-order ODE for p in terms of $\mathcal{L}$. Try the ansatz $p \propto e^{-\beta \mathcal{L}}$ and solve for β.

(b) The ratio $\frac{2}{\eta\sigma^2}$ plays the role of an inverse temperature β. Interpret what happens to the equilibrium distribution when: - η is very small (low temperature) - η is very large (high temperature)

Which regime favors flatter minima, and why might this be beneficial for generalization?

(c) In practice, SGD noise is **not** isotropic: $\Sigma(\theta)$ depends on both the loss landscape and the data. Without solving anything, explain qualitatively why anisotropic noise can introduce an implicit bias that goes beyond what the loss function $\mathcal{L}$ alone would select. Specifically, why might SGD preferentially escape sharp directions of the loss while remaining stable along flat directions?